

Branden Ghena

Contact Information

Computer Science Department
Northwestern University
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Employment

Associate Professor of Instruction, Northwestern University (2026–Present)
McCormick Advisory Council Professorship (2026–2029)
Department of Computer Science, McCormick School of Engineering
Evanston, Illinois

Assistant Professor of Instruction, Northwestern University (2020–2026)
Department of Computer Science, McCormick School of Engineering
Evanston, Illinois

Education

University of California, Berkeley, Berkeley, California (2017–2020)
PhD — Electrical Engineering and Computer Science
Dissertation: Investigating Low Energy Wireless Networks for the Internet of Things
Advisor: Prabal Dutta

University of Michigan, Ann Arbor, Ann Arbor, Michigan (2013–2017)
M.S.E. Computer Science and Engineering (2017)
PhD Candidate — Computer Science and Engineering
Advisor: Prabal Dutta

Michigan Technological University, Houghton, Michigan (2008–2013)
B.S. Computer Engineering & Electrical Engineering

Teaching Experience

Northwestern University, Evanston, Illinois

- Instructor, CS211: Fundamentals of Computer Programming II (W21,F21,W22,S23)
– Topics: C and C++ programming, memory management, and the Unix shell. 120–200 students.
- Instructor, CS213: Introduction to Computer Systems (F20,S21,W22,W23,F23,W24,W25,W26)
– Topics: data representation, control flow, caches, and parallelism. 150–250 students.
- Instructor, CS343: Operating Systems (F20,S22,F22,S24,F24,F25,F26)
– Topics: concurrency, scheduling, virtual memory, and file systems. 80–150 students.
- Instructor, CE346/CS346: Microcontroller System Design (S21,F21,F22,F23,F24,S25,F25,S26,F26)
– Topics: microcontrollers, communication buses, and embedded software. 50–60 students.

- Instructor, CS433: Wireless Protocols for the Internet of Things (W21,S22,W23,S24,S25,S26)
- Topics: Bluetooth Low-Energy, 802.15.4, and Low-Power Wide-Area Networks. 20–40 students.

University of California, Berkeley, Berkeley, California

- Instructor, CS61C: Great Ideas in Computer Architecture (Summer 2019)
- Topics include assembly, pipelined processor design, caches, and parallelism.
 - 200 students, 3 instructors, 8 TAs, 8 tutors
- Graduate Student Instructor, EE 149: Introduction to Embedded Systems (Fall 2018)
- Topics include embedded software, sensors, communication, and modeling cyber-physical systems.
 - 60 students, 2 instructors, 3 TAs
 - Redesigned lab curriculum to improve student learning and better prepare students for the open-ended course project. New labs use a simplified code development environment with Git, GCC, and Makefiles and a custom hardware platform, Buckler, built upon the nRF52832 dev kit.
 - Outstanding Graduate Student Instructor award
- Guest Lecturer, EE 149: Introduction to Embedded Systems (Fall 2017–2019)

University of Michigan, Ann Arbor, Ann Arbor, Michigan

- Graduate Student Instructor, EECS 370: Computer Organization (Fall 2013)
- Topics include C programming, assembly, pipelined processor design, and caches.
 - 350 students, 3 instructors, 8 TAs
 - Outstanding Graduate Student Instructor award
- Guest Lecturer, EECS 373: Design of Microprocessor-Based Systems (Fall 2014, Winter 2015)

Advising

Northwestern University, Evanston, Illinois

- Undergraduate Andrew Xue, “Development of a Formula Car Lap Counter” (S26)
- Undergraduates Natalie Hill and Jason Lu, “Development of a Teaching File System for CS343” (F24,W25)
- Undergraduates Anthony Alvarez and Jason Hu, “Tock: PIO SPI Driver” (F24,W25)
- Undergraduates Mary Caserio, Jason Lu, Randy Truong, and Mavis Wang, “Tock: Remote System Calls” (F23,W24)
- Undergraduate Joshua Feist, “Development of a Battery Monitoring System” (W22)

Research Experience

University of California, Berkeley, Berkeley, California

Graduate Student Research Assistant, Electrical Engineering & Computer Science (2017–2020)

University of Michigan, Ann Arbor, Ann Arbor, Michigan

Graduate Student Research Assistant, Computer Science and Engineering (2013–2017)

Michigan Technological University, Houghton, Michigan

Aerospace Enterprise: Designed and built a satellite with the Air Force Research Lab (2009–2013)

On-Board Data and Command Team Leader: Led team to complete computer hardware on the satellite,

including thermal, wiring, and acceptance testing (2012–2013)
University Nanosatellite Program: First place finish in national competition (2011).
Oculus-ASR satellite launched via Falcon Heavy on the STP-2 mission (2019).

NSF Research Experience for Undergraduates: Research in the field of Hybrid-Electric Vehicles. Studied analytical model for a continuously variable transmission. Created procedure for vehicle coast-down testing (2011)

NASA Jet Propulsion Laboratory, Pasadena, California

Summer Internship: Created testing framework for ASIC processor designs including both hardware and software testing (2013)

Summer Internship: Designed and fabricated a computer board to interface a thruster with a satellite Command and Data Handling system (2012)

Industry Experience

Hitachi Global Storage Technologies, Rochester, Minnesota

Test and Tools Co-op Student: Created hard drive testing tools using TCL and C++ languages. Worked with HDDs and SSDs as well as SATA and SCSI protocols (2010)

Fermi 2 Nuclear Power Plant, Monroe, Michigan

Summer Internship: Created plant configuration management documentation (2009)

Open Source Projects

Tock Embedded Operating System, <https://github.com/tock/tock>

Developer and Core Working Group team member (2015–Present)

Board Member of Tock Foundation (2025–Present)

Conference, and Journal Publications

1. Leon Schuermann, Bradford Campbell, **Branden Ghena**, Philip Levis, Amit Levy, and Pat Pannuto. Tock: From Research to Securing 10 Million Computers. In *Proceedings of the 31st Symposium on Operating Systems Principles, SOSP’25*, New York, NY, USA, Oct 2025. ACM. Acceptance Rate: 65 of 368
2. Nabeel Nasir, Viswajith Govinda Rajan, Pat Pannuto, **Branden Ghena**, and Bradford Campbell. Experiences teaching a wireless for the internet of things course cooperatively at multiple universities. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education, SIGCSE’24*, March 2024. Acceptance Rate: roughly 33%
3. Thomas Zachariah, Neal Jackson, **Branden Ghena**, and Prabal Dutta. Reliable: Towards reliable communication via bluetooth low energy advertisement networks. In *Proceedings of the 2022 International Conference on Embedded Wireless Systems and Networks, EWSN’22*, October 2022. Acceptance Rate: 14 of 46
4. **Branden Ghena**, Joshua Adkins, Longfei Shangguan, Kyle Jamieson, Phil Levis, and Prabal Dutta. Challenge: Unlicensed lpwans are not yet the path to ubiquitous connectivity. In *Proceedings of the 25th Annual International Conference on Mobile Computing and Networking, MobiCom’19*, October 2019. Acceptance Rate: 30 of 186
5. Joshua Adkins, **Branden Ghena**, Neal Jackson, Pat Pannuto, Samuel Rohrer, Bradford Campbell, and Prabal Dutta. The Signpost Platform for City-Scale Sensing. In *Proceedings of the 17th ACM/IEEE International Conference on Information Processing in Sensor Networks, IPSN’18*, New York, NY, USA, April 2018. ACM. Acceptance Rate: 22 of 83

6. Amit Levy, Bradford Campbell, **Branden Ghena**, Daniel B. Giffin, Pat Pannuto, Prabal Dutta, and Philip Levis. Multiprogramming a 64kB Computer Safely and Efficiently. In *Proceedings of the 26th Symposium on Operating Systems Principles, SOSP'17*, New York, NY, USA, Oct 2017. ACM. Acceptance Rate: 17%
7. Bradford Campbell, Meghan Clark, Samuel DeBruin, **Branden Ghena**, Neal Jackson, Ye-Sheng Kuo, and Prabal Dutta. Perpetual Sensing for the Built Environment. *IEEE Pervasive Computing*, 15(4), 2016
8. Samuel DeBruin, **Branden Ghena**, Ye-Sheng Kuo, and Prabal Dutta. PowerBlade: A Low-Profile, True-Power, Plug-Through Energy Meter. In *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems, SenSys'15*. ACM, 2015. Acceptance Rate: 27 of 132

Workshop Publications

1. **Branden Ghena**, Jean-Luc Watson, and Prabal Dutta. Embedded OSES must embrace distributed computing. In *Proceedings of the 1st International Workshop on Next-Generation Operating Systems for Cyber-Physical Systems, NGOSCPS'19*, April 2019
2. Joshua Adkins, **Branden Ghena**, and Prabal Dutta. Freeloader's Guide Through the Google Galaxy. In *Proceedings of the 20th International Workshop on Mobile Computing Systems and Applications, HotMobile'19*, New York, NY, USA, February 2019. ACM. Acceptance Rate: 26 of 57
3. Joshua Adkins, Bradford Campbell, **Branden Ghena**, Neal Jackson, Pat Pannuto, and Prabal Dutta. Energy Isolation Required for Multi-tenant Energy Harvesting Platforms. In *Proceedings of the Fifth ACM International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems, ENSys'17*, New York, NY, USA, November 2017. ACM. Acceptance Rate: 6 of 18
4. Amit Levy, Bradford Campbell, **Branden Ghena**, Pat Pannuto, Prabal Dutta, and Philip Levis. The Case for Writing a Kernel in Rust. In *Proceedings of the 8th Asia-Pacific Workshop on Systems, APSys'17*, New York, NY, USA, September 2017. ACM
5. Amit Levy, Michael P Andersen, Bradford Campbell, David Culler, Prabal Dutta, **Branden Ghena**, Philip Levis, and Pat Pannuto. Ownership is Theft: Experiences Building an Embedded OS in Rust. In *Proceedings of the 8th Workshop on Programming Languages and Operating Systems, PLOS* 2015. ACM, Oct 2015. Acceptance Rate: 7 of 16
6. Brad Campbell, **Branden Ghena**, and Prabal Dutta. Energy-Harvesting Thermoelectric Sensing for Unobtrusive Water and Appliance Metering. In *Proceedings of the 2nd International Workshop on Energy Neutral Sensing Systems, ENSys '14*. ACM, November 2014. Acceptance Rate: 9 of 11
7. **Branden Ghena**, William Beyer, Allen Hillaker, Jonathan Pevarnek, and J. Alex Halderman. Green Lights Forever: Analyzing the Security of Traffic Infrastructure. In *8th USENIX Workshop on Offensive Technologies, WOOT '14*. USENIX Association, August 2014. Acceptance Rate: 17 of 35

Magazine, Demo, and Poster Publications

1. Joshua Adkins, **Branden Ghena**, and Prabal Dutta. Signpost: Enabling City-Scale Sensing for Citizens and Scientists. In *GetMobile: Mobile Computing and Communications*, volume 22 of *GetMobile*, New York, NY, USA, September 2018. ACM
2. Joshua Adkins, Bradford Campbell, **Branden Ghena**, Neal Jackson, Pat Pannuto, Samuel Rohrer, and Prabal Dutta. Demo Abstract: Applications on the Signpost Platform for City-Scale Sensing. In *Proceedings of the 17th ACM/IEEE International Conference on Information Processing in Sensor Networks, IPSN'18*, New York, NY, USA, April 2018. ACM

3. Joshua Adkins, Bradford Campbell, **Branden Ghena**, Neal Jackson, Pat Pannuto, and Prabal Dutta. The Signpost Platform for City-Scale Sensing. In *TerraSwarm 2017 Annual Review*, TerraSwarm'17, October 2017
4. Bradford Campbell, **Branden Ghena**, Ye-Sheng Kuo, and Prabal Dutta. Demo Abstract: Swarm Gateway. In *Proceedings of the 3rd ACM International Conference on Systems for Energy-Efficient Built Environments*, BuildSys'16, November 2016
5. Joshua Adkins, Bradford Campbell, **Branden Ghena**, Neal Jackson, Pat Pannuto, and Prabal Dutta. Demo Abstract: The Signpost Network. In *Proceedings of the 14th ACM Conference on Embedded Networked Sensor Systems*, SenSys'16, November 2016
6. Bradford Campbell, **Branden Ghena**, Ye-Sheng Kuo, and Prabal Dutta. Demo Abstract: Swarm Gateway. In *Proceedings of the 3rd ACM International Conference on Systems for Energy-Efficient Built Environments*, BuildSys'16, November 2016
7. Samuel DeBruin, **Branden Ghena**, Ye-Sheng Kuo, and Prabal Dutta. Demo: PowerBlade A Low-Profile, True-Power, Plug-Through Energy Meter. In *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems*, SenSys'15. ACM, 2015

Dissertation

1. **Branden Ghena**. Investigating Low Energy Wireless Networks for the Internet of Things. In *University of California, Berkeley Dissertation*, 2020

Service

Northwestern Faculty Senate, McCormick Non-Tenure Representative (2025–Present)

Northwestern CS Study Abroad (2021–Present)

Northwestern TCG Club Advisor (2021–Present)

Northwestern CS Research Track Mentor (W24,W25)

Northwestern CE Tenure-track Faculty Search Committee (W23)

Northwestern CS Teaching-track Faculty Search Committee (W22)

PC Member, DATA Workshop at IPSN (2020–2022)

PC Member, LP-IoT Workshop at MobiCom (2021)

Awards and Grants

Cole-Higgins Award for Excellence in Teaching, Northwestern University, McCormick School of Engineering (2026)

McCormick Advisory Council Professorship (2026-2029)

NSF POSE Phase II Grant, Tock Secure Embedded Operating System, \$1,500,000 (2023-2026)

Cole-Higgins Award for Excellence in Teaching, Northwestern University, McCormick School of Engineering (2022)

Instructor of the Year, Northwestern University, Computer Science Department (2022)

Northwestern Associated Student Government Faculty & Administrator Honor Roll (2022)
Outstanding Graduate Student Instructor, UC Berkeley EECS Department (Fall 2018)
Best Demo Runner Up - IPSN'18 (2018)
David Wessel Best Demo Award - Terraswarm Annual Review (2017)
TI Innovation Challenge 2015 - Best Environmental Impact, \$2,000 (2015)
National Science Foundation Graduate Research Fellowship, \$96,000 plus tuition (2014–2017)
Outstanding Graduate Student Instructor, Michigan EECS Department (Fall 2013)
Michigan Technological University, Electrical and Computer Engineering Departmental Scholar (2012)

Other

Erdős Number: 4
(Branden Ghena - Alex Halderman - Ronald Rivest - David Avis - Paul Erdős)